



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Beyond the Viewer: Documentation Standards as Governance Infrastructure in 3D Heritage

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Abstract

Web-based 3D heritage is often discussed through the language of platforms, viewers, repositories, and access. Yet the long-term value of 3D heritage data depends less on visualisation alone than on the documentation structures that determine whether models can be interpreted, trusted, reused, and preserved. This paper argues that documentation standards should be understood not only as technical instruments for describing digital objects, but as governance infrastructure: systems that organise evidence, assign responsibility, stabilise institutional authority, and shape the future reuse of heritage data.

The paper develops a three-layer analytical model for understanding documentation standards in 3D heritage. The first layer is the data layer, where geometry, imagery, point clouds, meshes, textures, and derived models are produced and transformed through specific capture and processing workflows. The second is the metadata and paradata layer, where information about authorship, methods, uncertainty, processing decisions, source materials, software environments, and interpretive assumptions is recorded (Denard, 2009). The third is the governance layer, where institutional mechanisms such as validation, versioning, repository policy, access control, preservation responsibility, and community review determine how 3D data becomes durable, accountable, and interoperable.

This model is developed through two contextual case studies. The first examines CIPA/ICCROM-related photogrammetric documentation initiatives from the late twentieth century, showing how international cooperation around metric recording helped define documentation as both a technical and institutional practice (ICOMOS, 1996). The second examines the ArCH benchmark dataset for architectural heritage point-cloud segmentation, showing how machine-readable heritage datasets are governed through classification schemas, annotation protocols, benchmark criteria, versioned data structures, and validation procedures that make heritage objects available for computational reuse (Matrone et al., 2020). Together, these cases show that standards do not merely describe heritage data after it is produced; they actively configure what counts

as evidence, which forms of uncertainty remain visible, and how institutional responsibility is distributed across data creators, repositories, users, and future systems.

By connecting these historical and contemporary examples to current FAIR and TRUST-oriented debates (Wilkinson et al., 2016; Lin et al., 2020), the paper positions documentation standards within a wider infrastructural landscape that includes local university and library repositories that manage deposit, access, and preservation; national research-data initiatives such as the German NFDI and NFDI4Culture that coordinate standards, services, and community-specific data practices; and European/international infrastructures such as the European Collaborative Cloud for Cultural Heritage that seek to federate cultural-heritage data, tools, and expertise across institutions.

Its contribution is therefore both conceptual and practical. Conceptually, it reframes FAIR 3D heritage as a problem of governance, not only of technical interoperability. Operationally, the data layer supports FAIR findability and reuse through persistent, interpretable datasets; the metadata and paradata layer supports FAIR accessibility and interoperability by documenting provenance, methods, uncertainty, and transformation histories; and the governance layer aligns with TRUST principles by assigning responsibility for validation, version control, access policy, preservation, and institutional accountability. Practically, the paper identifies documentation requirements that should be made explicit in 3D workflows: provenance chains, paradata fields, transformation histories, versioning strategies, validation procedures, rights and access conditions, and preservation responsibilities. The paper concludes that sustainable 3D heritage infrastructures require documentation standards that operate “beyond the viewer”: as mechanisms of accountability, institutional memory, and long-term public trust.

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